# Watcharapon Thodraksa

Bangkok, Thailand, 10150 | Open to relocating internationally
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linkedin.com/in/watcharaphon-t portfolio-watcharapon.vercel.app github.com/watcharaponthod-code

## SKILLS

**Programming Languages:** Python, SQL, TypeScript, JavaScript, Java

**ML / Computer Vision:** PyTorch, OpenCV, semantic segmentation (LR-ASPP, DeepLab), DINOv2, EfficientNet, YOLO (detection, oriented boxes), SAM 2, CLIPSeg, optical flow, audio event detection, scikit-learn, LightGBM

**Training / Optimization:** knowledge distillation, hard-negative mining, self-training with pseudo-labels, synthetic data, score calibration, ONNX, OpenVINO

**MLOps/DevOps:** Kubernetes (k3s), Kustomize, Helm, Docker, MLflow, Flux GitOps, GitHub Actions, GitLab CI, FastAPI, pytest, Grafana, Prometheus, Kaggle GPU, Railway

**LLM / Backend:** LangGraph, LangChain, pgvector, BGE-M3, Ollama, RAG, Spring Boot, Kafka, PostgreSQL

## EDUCATION

**Kasetsart University**

Bachelor of Science (B.Sc.) | Major: Computer Science (Co-op Program)

## PROFESSIONAL EXPERIENCE

**AI/ML Engineer** **TRR** 2026 – Present

Computer vision and satellite ML for a sugar mill group in Thailand.

- **CaneSight:** CCTV system that scores every truck dump for dust, leafy cane and burnt cane, and sends the result to the mill's app. Runs on a CPU-only k3s server inside the mill.
- **MLOps:** MLflow registry on k3s, gated training on Kaggle, automatic deploy with rollback, Flux GitOps, daily drift monitoring.
- **Burnt-cane classifier:** AUC 0.994 on 483 trucks from days the model never saw (labels from 25,980 weighbridge tickets).
- **CropScan:** harvest detection from Sentinel-1/2 satellite images; F1 0.883 on 8,924 held-out fields.
- **Yield Pro:** rebuilds the crop signal from radar when clouds block the image; 95.7% of values within ±0.10.
- **CaneGate:** truck inspection at the weighbridge with Thai plate OCR, shipped as Docker images, used at the mill.

**DevOps & Backend Engineer (Intern)** **Sycapt** Jan 2026 – May 2026

- Built an on-premises RAG system (LangGraph, BGE-M3, pgvector, Ollama) on Kubernetes for a network with no internet access.
- Ran four LLMs on one 16 GB GPU by swapping models through RAM.
- Ran Kubernetes and Helm deployments, the GitLab CI/CD pipeline with automatic rollback, and Grafana/Prometheus monitoring.

**Programmer (Intern)** **Sycapt** Nov 2025 – Jan 2026

- Developed backend services for a payment provider handling 1M+ transactions per day (PromptPay, Alipay).
- Built a Kafka connector microservice (Java, Spring Boot) and a map dashboard for 15,423 payment terminals.

## PROJECTS

**Portfolio website:** portfolio-watcharapon.vercel.app (case studies for the work above)

**YouTube Shorts Automation:**

*Python · YouTube Analytics API · ffmpeg*

- Pipeline that reads channel analytics, picks topics, generates and uploads clips on schedule; 3.34M views across 189 clips.

**Teaching and Awards:**

- Invited lecturer at Kasetsart University: workshops on React Native with the Gemini API and on Git/GitHub.
- 1st Runner-up, KUSE AI Hackathon 2025 · Winner, UI Design-athon 2025.